Appendix 23

Density matrices in the lowest Landau level

In this appendix we show that the one-particle density matrix of a many-body state in the lowest Landau level can be explicitly computed from the electronic density of that state (MacDonald and Girvin, 1988). The reason for this surprising result lies in the special analytic structure of the wave functions in the lowest Landau level. Recall that, according to Eq. (10.41), the one-particle density matrix

$$\rho(\vec{r},\vec{r}\,') = \int \psi^*(\vec{r},\vec{r}_2\ldots\vec{r}_N)\psi(\vec{r}\,',\vec{r}_2\ldots\vec{r}_N)d\vec{r}_2\ldots d\vec{r}_N \tag{A23.1}$$

can be written in the form

$$\rho(z,z') = e^{-\frac{|z|^2+|z'|^2}{4\ell^2}}F(z^*,z')\,, \tag{A23.2}$$

where z and z' are the complex variables associated with the two-dimensional vectors $\vec{r}$ and $\vec{r}\,'$, and the function F is analytic in each of its two arguments. The density, in this representation, is given by

$$n(\vec{r}) = \rho(\vec{r},\vec{r}) = e^{-\frac{z^*z}{2\ell^2}}F(z^*,z)\,. \tag{A23.3}$$

Now observe that an arbitrary function of position $n(\vec{r}) = n(x,y)$ can be represented as a function of z and z^* simply by substituting for the coordinates x and y their expressions $x = \frac{z+z^*}{2}$ and $y = \frac{z-z^*}{2i}$.

Comparing this with Eq. (A23.3) we immediately see that

$$F(z^*,z) = e^{\frac{z^*z}{2\ell^2}}n\left(\frac{z+z^*}{2},\frac{z-z^*}{2i}\right)\,. \tag{A23.4}$$

The key point is that this equation completely determines the analytic dependence of $F(z^*,z)$ on each of its two arguments. By varying these arguments independently we arrive at the identification

$$F(z^*,z') = e^{\frac{z^*z'}{2\ell^2}}n\left(\frac{z'+z^*}{2},\frac{z'-z^*}{2i}\right)\,. \tag{A23.5}$$

Finally, by substituting this formula in Eq. (A23.2) we obtain the one-particle density matrix

$$\rho(z,z') = e^{-\frac{|z|^2+|z'|^2-2z^*z'}{4\ell^2}}n\left(\frac{z'+z^*}{2},\frac{z'-z^*}{2i}\right)\,. \tag{A23.6}$$

In the special case of a state of uniform density, Eq. (A23.6) tells us that the density matrix reduces to the simple form

$$\rho(z,z') = n e^{-\frac{|z-z'|^2}{4\ell^2}} e^{\frac{z^* z' - z z'^*}{4\ell^2}} \;, \quad \text{(uniform density)} \;, \tag{A23.7}$$

which tends to zero exponentially for $|z - z'| \to \infty$.

Eq. (A23.6) is used in Exercise 10 to obtain the explicit expression of the Hartree–Fock potential in the LLL in terms of the projected density operator. A similar analysis allows the construction of the two-particle density matrix in terms of the pair correlation function.